

Roll No.

DETF-05

B. TECH. (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
INTERNAL COMBUSTION ENGINE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) How are the heat engines classified ? Also, classify the internal combustion engine in detail. (CO1)

OR

- (b) What are the important basic components of an IC engine ? Explain them briefly. (CO1)

2. (a) Explain the working of a four-stroke CI engine with the help of a neat sketch. (CO1)

OR

- (b) Compare the four-stroke engine and two-stroke engine. Also describe the various parameters used to evaluate the engine performance. (CO1)

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(2)

3. (a) What is the difference between air-standard cycle and fuel-air cycle ?
What are the assumptions used in fuel-air cycle ? Explain taking the
example of Otto cycle. (CO1)

OR

- (b) Explain the following : (CO1)
- (i) Variation in specific heats with temperature
 - (ii) Dissociation
4. (a) What do you mean by carburetion and air-fuel mixtures ? Also explain
about the mixture requirements at different loads and speeds. (CO2)

OR

- (b) Explain the working of a simple carburetor with the help of a neat
sketch. (CO2)
5. (a) Write the functional requirements of an injection system. (CO2)

OR

- (b) Classify the fuel injection system and explain each type in detail. (CO2)